CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)

Nov/Dec 2012

EE310 Programmable Logic Controller and Industrial Automation

Date: 09.11.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Draw and explain block diagram of PLC in detail [06]

(b) List the types of counters available in PLC [01]

Q - 2 (a) Draw and explain working of ON delay timer (along with timing diagram). Write a program in PLC for flashing of a light for 1sec. (Light will go on for 1sec after that will turn off for 1 sec, and the process will go on.) [12]

(b) State function of input module. [02]

OR

Q - 2 (a) Draw and explain working of OFF delay timer (along with timing diagram). [06]

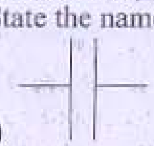
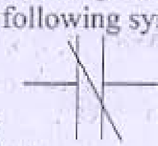
(b) Three conveyors feed a main conveyor. The count from each feeder conveyor is fed into an input register in PLC. Construct a PLC program to obtain the total count of parts on the main conveyor. [06]

(c) State function of output module. [02]

Q - 3 (a) With the help of necessary figure explain in detail scan cycle of PLC [06]

(b) Make ladder diagram for the following sequence: [06]
When SW1 is closed, CR1 goes on
After CR1 goes on, SW2 can turn CR2 on
When CR2 goes on, PL1 goes off

(c) State the name of following symbols: [02]

i)  ii) 

OR

Q - 3 (a) Implement logic gates in PLC in ladder logic. (AND, OR, NOT, NOR, NAND, EXOR and EXNOR) [14]

SECTION – II

Q - 4 (a) Do as directed: [07]

i. Define transducer

ii. Define sensor
iii. State the range of standard current signal used for transmission in PLC
iv. Battery in the PLC is used for ___.
v. What is the difference between, sink and source module?
vi. What sort of voltages are generally used for PLC input's and output's?
vii. What is the basic difference between a PLC and relay control system?

Q - 5 (a) State the advantages and disadvantages of PLC system **[10]**

(b) When a switch is turned on, C goes on immediately and D goes on 9 seconds later. Opening the switch turns both C and D off. **[04]**

OR

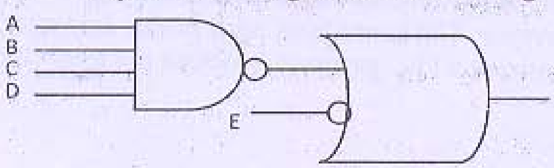
Q - 5 (a) For the following questions below convert the word description to **gate symbols** and **PLC ladder logic diagram:** **[09]**

i. Switch 8 and switch 11, plus either switch 22 or switch 34, must be on for output 67 to be on.
ii. For output 7 to be on, input 6 must be off and either input 8 or input 9 must be on. In addition, one of inputs 1, 2 or 3 must be on.
iii. For output H to be on and both inputs C and D must be off. In addition one or more inputs E, F and G must be off.

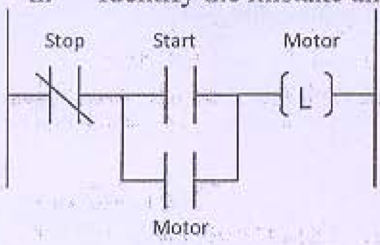
(b) List the types of programming languages used to program a PLC. Write two lines about each one of them. **[05]**

Q - 6 (a) Do as directed: **[06]**

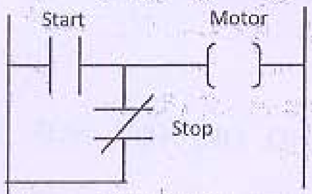
i. Prepare ladder logic for the following:



ii. Identify the mistake and rectify it.



iii. Prepare gate symbol for the following if possible



iv. With the increase in the length of wire current ___.
v. What is the full form of HMI
vi. Which of the following is not normally in the array of status indicators found on the processor module:
   a) Memory OK

b) Wiring OK
c) Battery OK

(b) There are two conveyors, each with sensors to count input and output of parts entering and leaving the conveyors. Prepare a ladder logic to glow three indicating lights for the process as follows: **[06]**
Number of parts on the both the conveyors equal - White light
Number of parts on conveyor 1 greater than conveyor 2 – Green light
Number of parts on conveyor 2 greater than conveyor 1 – Blue light

(c) Prepare a ladder diagram for an output is on if the input count is less than 34 or more than 41. **[2]**

OR

Q - 6 (a) Multiple Choice questions: **[05]**

1. What do the letters 'AQ' normally stand for in PLC addresses?
   a) Anything Quaint
   b) Analog out of Range
   c) Analog Output
   d) Nothing, it is no typical or common use in PLC's
2. What does 'X' normally stand for in PLC addresses?
   a) Timer
   b) Counter
   c) Digital Output
   d) Digital Input
3. What does 'Q' normally stand for in PLC addresses?
   a) Timer
   b) Digital Input
   c) Counter
   d) Digital Output
4. What time delay do a timer preset of 33 & a time base of 0.01 give you?
   a) 33 seconds
   b) 0.33 seconds
   c) 3.3 seconds
   d) Nothing, this question doesn't make any sense
5. PLC's does a self-check every scan cycle.
   a) True
   b) False

(b) A machine M is to be turned on either when a count A goes up to 11 or when count B goes up to 16. One stop button or switch resets the entire process. **[05]**

(c) A fan is to be started and stopped from any one of the three locations. Each location has a start and a stop button. Prepare ladder logic for the same. **[04]**

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